

Program 8: Create Kubernetes Pods using YAML Descriptors

Project Structure

program8

- |--nodejs-configmap.yaml
- |--nodejs-pod.yaml
- |--[server.js](#)

Start the minikube

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg8$ minikube start --driver=docker
🐹 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚧 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
❗ Failing to connect to https://registry.k8s.io/ from both inside the minikube container and host machine
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
```

Step1:

Create a node js app

Nano [server.js](#)

```
GNU nano 7.2 server.js
const http=require("http")
const port=3000;
const server=http.createServer((req,res)=>{
    res.end("Nodejs is running inside kubernetes");
});
server.listen(port,()=>{
console.log(`Server running on port {port}`);
});
```

Step2: create a nodejs-configmap.yaml file

```

GNU nano 7.2 nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http =require("http");
    const port = 3000;
    const server=http.createServer((req,res)=>{
      res.end(" Hello from Nodejs running inside kubernetes");
    });
    server.listen(port,()=>{
      console.log(`Server running at port ${port}`);
    });

```

Screenshot captured
You can paste the image

Step 3: create a nodejs-pods.yaml file

```

GNU nano 7.2 nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18-alpine
      command: ["node","/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config

```

Step4: Apply the yaml files

```

1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg8$ kubectl apply -f nodejs-pod.yaml
pod/program8 created

```

```

1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created

```

Step6: forward the port and acces the nodejs app

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg8$ kubectl port-forward pod/program8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

Step7: check the output on <http://localhost:3000>

